Exp: 8 Nmap Live Host Discovery.
28/9/22 ARP, ICMP, and TCP/UDP Ping Scans

Aim:

To discover live hosts on a network using Nmap host-discovery scans - ARP scan, ICMP scan and TCP/UDP ping scan - and to understand which protocol operates at which layer of TCP/IP model, and when each one applies.

ARP Host Discovery

```
$ sudo nmap -PR -sn 10.200.6.0/24
Starting Nmap 7.80 (https://nmap.org)
Nmap scan report for 10.200.6.1
Host is up.
Nmap done 256 IP addresses (1 host up) scanned in 10.51 seconds
```

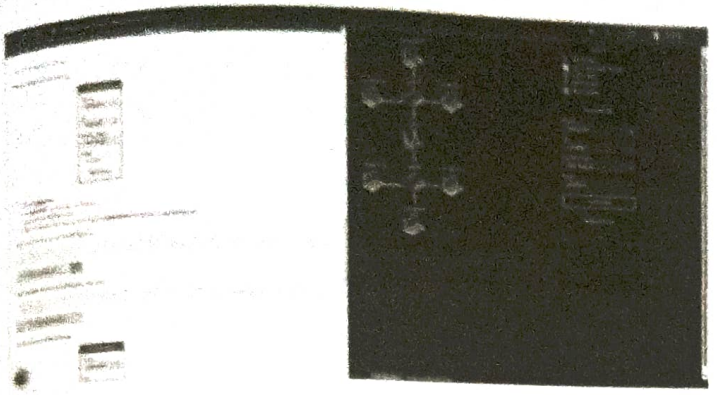
ICMP Host Discovery

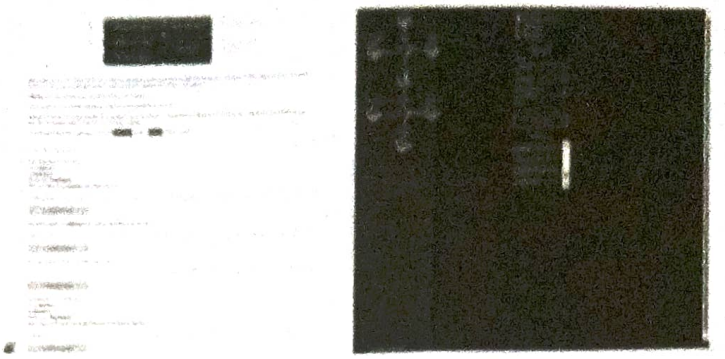
ICMP Echo (-PE)

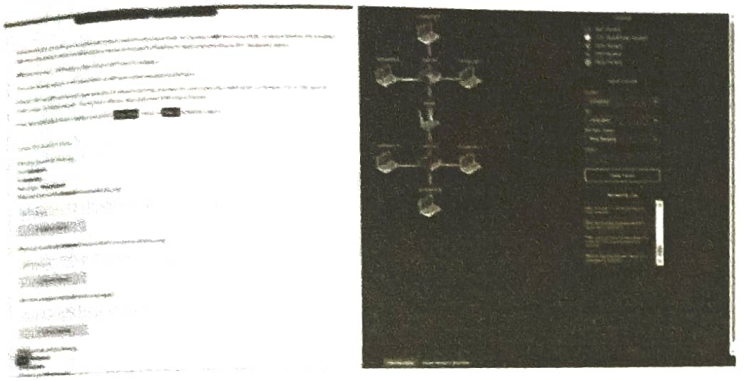
```
$ sudo nmap -PE -sn 10.200.6.0/24
Nmap scan report for 10.200.6.1
Host is up (0.069s latency)
Nmap scan report for 10.200.6.50
Host is up (0.070s latency)
Nmap scan report for 200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP addresses (3 hosts up) scanned in 3.38 seconds
```

ICMP Timestamp (-PP)

```
$ nmap -PP -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.069s latency).
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done. 256 IP addresses (2 hosts up) scanned in 8.44 seconds.
```



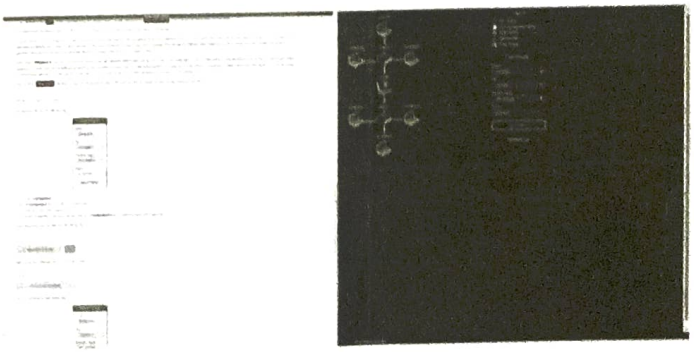




```
pentester@TryHackMe$ sudo nmap -PE -sn 10.10.68.220/24

Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 12:10 EEST
Nmap scan report for 10.10.68.50
Host is up (0.12s latency).
Nmap scan report for 10.10.68.52
Host is up (0.12s latency).
Nmap scan report for 10.10.68.77
Host is up (0.11s latency).
Nmap scan report for 10.10.68.110
Host is up (0.11s latency).
Nmap scan report for 10.10.68.140
Host is up (0.11s latency).
Nmap scan report for 10.10.68.142
Host is up (0.11s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap scan report for 10.10.68.222
Host is up (0.11s latency).
Nmap done: 256 IP addresses (8 hosts up) scanned in 8.26 seconds
```

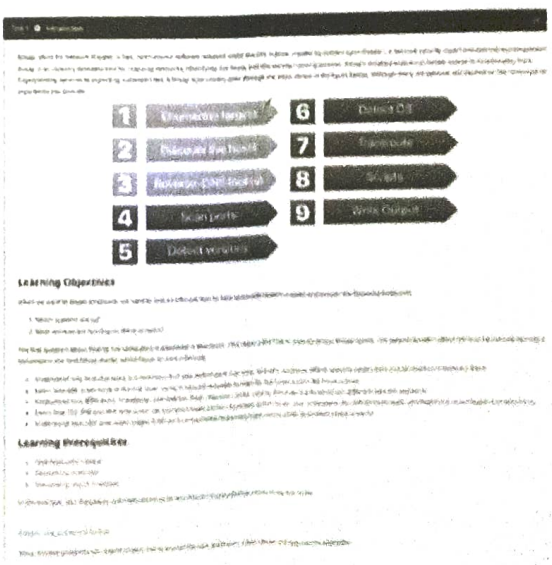




```
pentester@TryHackMe$ sudo nmap -PR -sn 10.10.210.6/24

Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-02 07:12 BST
Nmap scan report for ip-10-10-210-75.eu-west-1.compute.internal (10.10.210.75)
Host is up (0.00013s latency).
MAC Address: 02:83:75:3A:F2:89 (Unknown)
Nmap scan report for ip-10-10-210-100.eu-west-1.compute.internal (10.10.210.100)
Host is up (-0.10s latency).
MAC Address: 02:63:D0:1B:20:CD (Unknown)
Nmap scan report for ip-10-10-210-165.eu-west-1.compute.internal (10.10.210.165)
Host is up (0.00025s latency).
MAC Address: 02:59:79:4F:17:B7 (Unknown)
Nmap scan report for ip-10-10-210-6.eu-west-1.compute.internal (10.10.210.6)
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.12 seconds
```

```
pentester@TryHackMe$ sudo nmap -PU -sn 10.10.68.220/24
Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 13:45 EEST
Nmap scan report for 10.10.68.52
Host is up (0.10s latency).
Nmap scan report for 10.10.68.121
Host is up (0.10s latency).
Nmap scan report for 10.10.68.125
Host is up (0.14s latency).
Nmap scan report for 10.10.68.134
Host is up (0.096s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap done: 256 IP addresses (5 hosts up) scanned in 9.20 seconds
```



ICMP Address Mask (-PM)

$ nmap -PM -sn 10.200.6.0/24
Nmap done 256 IP addresses (0 hosts up) scanned in 52.14 seconds.

TCP and UDP Ping Discovery

TCP SYN Ping (-PS)

$ nmap -PS -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.010s latency)
Nmap scan report for 10.200.6.250
Host is up (0.068s latency)
Nmap done: 256 IP addresses (2 hosts up) scanned in 36.64 seconds

TCP ACK Ping (-PA)

$ sudo nmap -PA -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP addresses (2 hosts up) scanned in 36.64 seconds

UDP Pind (-PU)

$ sudo nmap -PU -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.010s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP address (2 hosts up) scanned in 35.97 seconds

Nascan - A Faster Alternative

$ nascan 10.200.6.0/24 -p443
$ nascan 10.200.6.0/24 -P80,443
$ nascan 10.200.6.0/24 -p22-25

Result :

Live hosts in the given network were simultaneously successfully identified using different nmap.